

Pavan Kumar Etta

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EDUCATION

University of Massachusetts, Dartmouth

Master of Science - Computer Information Systems, Data Science, GPA: 4.0

North Dartmouth, MA

Aug 2024 – May 2026

Indian Institute of Information Technology, Sricity

Bachelor of Technology - Electronics and Communication Engineering, GPA: 3.2

Chittoor, India

Aug 2017 – May 2021

SKILLS

Programming Languages: Python, Java, C++, C#, Go, JavaScript, TypeScript

Backend & Distributed Systems: Microservices, Distributed Systems, REST APIs, Event-Driven Systems

Databases & Storage: MySQL, PostgreSQL, MongoDB, Microsoft SQL Server (MSSQL), DynamoDB, Redis

Cloud & Infrastructure: AWS (Lambda, EC2, API Gateway, S3, Redshift), Docker, Kubernetes, Microsoft Azure

Data Engineering & Distributed Processing: Apache Spark, PySpark, Hadoop, Stream Processing, Data Pipelines

Frameworks: Node.js, React, React Native, Angular, Next.js, Vue.js, Django, Flask, GraphQL, Redux, HTML, CSS/SCSS

Machine Learning & Data Science: NumPy, Pandas, Scikit-learn, OpenCV, PyTorch, TensorFlow, NLTK

DevOps & Tools: GitLab CI/CD, Kafka, Linux, Bash, HTTP, TCP/IP, JIRA, Jest, AJAX, jQuery

PROFESSIONAL EXPERIENCE

Software Engineer II | Rhenix Lifesciences LLP

Aug 2021 – Aug 2024

- Designed a microservices-based full-stack architecture using Python, Django, and AWS (Lambda, API Gateway, EC2, DynamoDB) to process large-scale wearable telemetry and enable scalable, production-ready systems.
- Built and deployed full-stack applications and backend services, integrating REST APIs for mobile applications on Google Play Store and Apple App Store, supporting **2,000+** active users and improving system accessibility.
- Improved data ingestion throughput by **~35%** by building high-throughput REST APIs and distributed pipelines integrating Fitbit, Apple Health, and Withings APIs, ensuring reliable and scalable data processing.
- Architected event-driven pipelines using AWS MSK (Kafka) to enable real-time, asynchronous data workflows, improving system scalability and fault tolerance through multi-partition streaming design.
- Reduced system latency by **~40%** by implementing fault-tolerant distributed ingestion pipelines with scheduling, retries, and optimized workflows, enabling near real-time analytics.
- Developed a wearable data analytics dashboard processing 50K+ daily health data points, reducing preprocessing time by **40%**, and improving data availability for downstream insights.

Graduate Research Assistant | University of Massachusetts, Dartmouth

Nov 2024 – Dec 2025

- Developed React and Angular web apps with Typescript to present medical research data. Implemented features for data querying and visualization using Plotly.js and backend powered by Django and Python.
- Optimized the MySQL database schema, employing partitioning, and indexing for efficient storage and retrieval. Integrated a caching layer using Redis reducing database load and improving query response time by **70%**.
- Utilized Docker to containerize the app with enhanced scalability, leading to a **25%** improvement in active users.

Lead Software Engineer | Cure Science Institute

Jun 2021 – Aug 2021

- Built real-time telemetry ingestion pipelines using Python and NumPy, processing sensor streams from embedded hardware system devices.
- Implemented signal preprocessing and feature extraction pipelines using Python and MATLAB, improving signal quality and system stability by **~30%**.
- Developed machine learning based anomaly detection systems using Python and Scikit-learn, achieving **~93%** accuracy in equipment monitoring tasks.

Software Developer Intern | Rashtriya Ispat Nigam Limited

May 2019 – Jun 2019

- Analyzed large-scale industrial telemetry datasets using Python data processing pipelines, improving fault detection accuracy by **~20%**.
- Built preprocessing pipelines to clean and standardize high-volume sensor data for downstream analytics systems.

PROJECTS

Water Quality Analysis

(NetCDF, Tableau, PyTorch, PySpark, Hadoop, GCP)

- Processed **30GB** environmental datasets using PySpark and Hadoop to analyze water distribution trends and train LSTM models with an MAE of 1.45.

Fault-tolerant sharded key-value store

(C++, Raft, SQLite)

- Implemented a distributed key-value storage system using the **Raft** consensus protocol, enabling sharded replication and fault-tolerant data storage across distributed nodes.

GPU Power Consumption Analysis

(PyTorch, TensorRT, nvidia-smi)

- Developed algorithms to analyze GPU workload energy consumption using **PyTorch** and **TensorRT**, optimizing CPU/GPU configurations for improved efficiency in edge computing workloads.